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11 September 1963
Superseding
MIL-P-26795A(USAF)
30 December 1959

MILITARY SPECIFICATION

PUMP, ROTARY, POWER DRIVEN, 20 GPM CAP., TWO WHEELED HAND CART, TYPE MC-1

1. SCOPE

1.1 This specification covers one type of two-wheeled hand cart, 20 gpm capacity, power driven, extinguishing agent transfer rotary pump, designated Type MC-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed And Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-T-60	Tape, Pressure Sensitive, Waterproof - For Packaging and Sealing

Military

JAN-P-100	Packaging and Packing For Overseas Shipment, General Specification
MIL-P-116	Preservation, Methods of
MIL-B-138	Boxes, Wood, Fiberboard-Lined For Overseas Shipment (For Weight Contents Not Exceeding 500 Pounds)
MIL-T-152	Treatment, Moisture - And Fungus-Resistant, Of Communications, Electronic, And Associated Electrical Equipment

FSC 4320

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JAN-C-266	Charges, Fire-Extinguisher (Foam, Mechanical) Type 5
MIL-C-3432	Cable and Wires, Electrical (Power and Control, Flexible and Extra Flexible, 300 and 600 Volts)
MIL-B-4394	Bromochloromethane, Technical
MIL-H-4536	Hose, Rubber, Bromochloromethane Resistant
MIL-W-8005	Wheels and Hubs, for Industrial Pneumatic Tires
MIL-T-9107	Test Reports, Preparation of
MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible
MIL-M-26512	Maintainability Requirements for Aerospace Systems and Equipment
MIL-I-26600	Interference Control Requirements, Aeronautical Equipment
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-129	Marking For Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-808	Finishes, Protective, and Codes for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MS24217	Cable Assembly - Power, Electrical, 28 Volt DC Power
MS33586	Metals, Definition of Dissimilar
MS51336	Lunette-Coupler, Drawbar, Ring

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publication.- The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Handbook H28 Screw Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Preproduction.- This specification makes provisions for preproduction testing.

3.2 Components.- The pumping unit (see 6.3) shall consist of the following:

Item No.	Quantity	Item Name	See Requirement
1	1	Pump	3.7
2	2	Hose	3.8
3	1	Pump relief valve	3.9
4	1	Motor	3.10
5	1	Hand cart	3.11.

3.3 Selection of specifications and standards.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Protective treatment.- When materials are used in the construction of the pumping unit that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

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3.4.2 Dissimilar metals.- Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other (particularly brass, copper, or steel in contact with aluminum or aluminum alloy). However, metal plating or metal spraying of dissimilar base metals to provide similar or suitable butting surfaces will be permitted. The use of dissimilar metals separated by suitable insulating material will also be permitted. Dissimilar metals are as defined in MS33586.

3.5 Design and construction.- The pumping unit shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, and service.

3.5.1 All components shall be waterproofed and so designed that their normal functioning will not be impaired by entrance of casual water during heavy rains, as formed by condensation or resulting from spillage of extinguishing agents. The components shall also be designed to effectively resist any harmful effects of dust and fungi. The pumping unit shall be fabricated of materials that will resist the corrosive effects of bromochloromethane (CB) and are suitable for use with foam liquids.

3.5.2 Reliability.- The pumping unit shall have a minimum MTBF of 280 hours with a minimum reliability of 0.99 for a 0.2-hour mission at a confidence factor of 0.90 (see 6.3).

3.5.3 Maintainability.- Maintainability requirements shall be in accordance with MIL-M-26512.

3.5.4 Lubrication.- Where practicable, permanently lubricated (lubricated-for-life) components, or units not requiring lubrication, shall be utilized. Design of grease and oil seals shall provide maximum accessibility for replacement. Where feasible, all components subject to wear shall be properly lubricated and renewable. All lubrication fittings shall be readily accessible for servicing.

3.5.5 Guards and cover plates.- Dangerously exposed moving parts shall be protected with guards. Protective covers shall be placed over all parts of the pumping unit where the need exists for protection against weather, dirt, or agent corrosive action.

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3.6 Performance.- The pumping unit shall be capable of operating satisfactorily in either a vertical or horizontal position under the following conditions:

a. Storage temperatures ranging from -80° to +160° F and operating temperatures ranging from -65° to +125° F as follows:

(1) Pumping foam liquid (JAN-C-266) at a rate of not less than 20 gpm within a range of 32° to 125° F

(2) Pumping CB (MIL-B-4394) at a rate of not less than 13 gpm within a range of 60° to 100° F, and not less than 18 gpm within a range of -65° to +60° F

b. Dry operation for a period of 1 hour under a suction lift of 10 inches Hg

c. Relative humidity up to 100 percent including conditions wherein condensation takes place in the form of both water and frost

d. Exposure to salt sea atmosphere.

3.6.1 The pumping unit shall be towable at speeds up to 15 mph over a level, paved surface.

3.7 Pump.- The pump shall be of the positive-displacement- or self-priming centrifugal type, and shall be directly coupled to the motor. The pump shall be of adequate capacity to meet the performance requirements specified herein and shall be designed to withstand a static test case pressure of 20 psi. The intake and discharge ports of the pump shall be provided with fittings for attaching the intake and discharge hoses.

3.8 Hose.- All hose shall be of 1-inch diameter and shall conform to MIL-H-4536.

3.8.1 Intake hose.- The intake hose shall be 15 feet in length and shall be provided with a pickup tube to be inserted through the 1-1/2-inch diameter bunghole of either a 20- or 55-gallon shipping drum. A 10-mesh screen shall be provided in the pickup tube to prevent foreign matter from being drawn through the pump. The screen shall not interfere with the normal operation of the pump. Corrosion-resistant spring wire shall be fitted on the outside of the hose at both ends to eliminate kinking of the hose at the couplings. The spring wire shall be at least 12 inches long and held in such a manner to assure that the spring cannot pull away from the coupling.

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3.8.2 Discharge hose.- The discharge hose shall be 15 feet in length. It shall be used to connect to the fill line of agent tanks on aircraft fire-fighting trucks. The spring shall be as specified in 3.8.1, except it shall be fitted on the pump side only. The pump end of the hose shall be provided with a sight glass to assist in determining when flow of agent is obtained. The sight glass shall be equipped with a petcock to permit elimination of air from the system. When specified (see 6.2), the following couplings shall be provided and fitted on the delivery end of the hose:

a. Quick-disconnect, self-sealing type coupling with wingnut, (Aeroquip Part No. 5100-S5-16, or equal) when used with agent conforming to MIL-B-4394

b. Coupling, 1-1/2 inch NH external thread, in accordance with Handbook H28, when used with agent conforming to JAN-C-266.

3.9 Pump relief valve.- A bypass pressure-relief valve shall be provided on the pump to assure against damage to the system in the event of valve failure in the agent system.

3.10 Motor.- A 28-V d-c compound wound, totally enclosed, weatherproof motor shall be coupled to the pump. The motor shall be of the ball bearing type and shall be provided with A-J tropical insulation treatment in accordance with MIL-T-152. The temperature rise of the motor shall not exceed 130° F on continuous operation under rated load conditions.

3.10.1 Power cable.- The pumping unit shall be provided with a 20-foot, 2 No. 12 AWG electric cable conforming to MIL-C-3432, class HD and having an MS24217 connector head molded to the cable.

3.10.2 Motor control cable.- The motor shall be controlled by a double-pole, single-throw momentary contact, normally off switch mounted at the end of a 15-foot cable conforming to MIL-C-3432, class HD. The switch shall be totally weatherproof and shall be of such design that it may be operated by personnel wearing heavy arctic mittens.

3.11 Hand cart.- A two-wheel hand cart shall be provided for easy transportation of the pumping unit which shall be permanently bolted to the cart. The cart shall have pneumatic tires, wheels, and hubs of at least 3.50-6 size conforming to MIL-W-8005. The handle of the cart shall be provided with a lunette coupler conforming to MS51336 to facilitate transportation when connected to the pintle hook of a vehicle.

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3.11.1 Hose and electric cord storage.- Racks shall be provided on the hand cart for the storage of all hose and electric cord. The storage racks shall be so located and designed that the hoses and cords are easily removed and interference is avoided between the ground and components when the cart is in either a vertical or horizontal position.

3.12 Part numbering of interchangeable parts.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-70327 shall govern the manufacturer's part numbers and changes thereto.

3.13 Electromagnetic interference.- The pumping unit shall comply with the requirements of MIL-I-26600, class IV.

3.14 Dimensions

3.14.1 Height.- With the cart in the horizontal position (lunette coupler on the ground), the overall height of the pumping unit shall not exceed 24 inches.

3.14.2 Length.- The overall length of the cart shall not exceed 48 inches.

3.14.3 Width.- The width of the pumping unit shall not exceed 25 inches.

3.15 Weight.- The entire assembly, including all required hose, electric cable and connections, pump, motor, hand cart, and pickup tube, shall not exceed 215 pounds. The weight of the unit shall be so distributed that 85 +5 -0 percent will be on the main wheels and the remaining weight will be on the lunette coupler while the unit is in a towing position with the lunette coupler 20 +10 -0 inches above the ground.

3.16 Finishes and protective coatings

3.16.1 Cleaning, painting, plating, anodic films, and chemical treatments shall be in accordance with MIL-STD-808. When used, aluminum alloys shall have exterior and interior surfaces anodically treated, and steel parts that come into direct contact with the agent, with the exception of stainless steel, shall have exterior and interior surfaces cadmium plated. Insignia red paint, film designation BG, shall be used on all parts of equipment normally painted in good commercial practice.

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3.17 Identification of product.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.18 Workmanship.- The pumping unit, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs and sharp edges; marking of parts and assemblies; thoroughness of soldering, welding, brazing, plating, wiring, and riveting; alignment of parts and tightness of assembly screws and bolts; et cetera.

3.18.1 Castings.- Castings shall be of high-grade quality, clean, sound, and free from blow holes, porosity, cracks, and any other defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests.- The inspection and testing of the pumping unit shall be classified as follows:

a. Preproduction testing. . . . . See 4.4

b. Acceptance tests . . . . . See 4.5.

4.3 Test conditions

4.3.1 Agent.- Agent conforming to JAN-C-266 and MIL-B-4394 shall be used during the tests, except that foam liquid shall not be pumped during the low temperature test.

4.3.2 Preparation for tests.- The pumping unit submitted for test shall be properly prepared for operation. The pumping unit shall be in correct adjustment, lubricated, and serviced for immediate operation.

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4.3.3 Data.- During the testing specified herein, at least the following data shall be recorded on log sheets at 1/4-hour intervals, except where shorter intervals are a test requirement. This data shall be included in the test report.

Date _____

Time _____ hours

Ambient temperature _____ F

Head _____ feet of CB (foam) or _____ psi

Agent delivery rate gpm _____

Agent temperature _____

Potential _____ volts

Current _____ amperes.

4.3.3.1 Notes shall be placed on the log sheets of all incidents of the tests, such as adjustments made, servicing performed, leaks, vibrations, electrical shorts, and any other irregular functioning of the pumping unit or its components and the corrective measures taken.

4.3.4 Test time.- Intervals of endurance test time less than 1 hour terminated by a parts failure shall not be credited to the endurance test. Endurance test time shall be credited in 1/2-hour intervals.

4.4 Preproduction testing

4.4.1 Test sample.- One pumping unit shall be subjected to the tests specified in 4.4.3.

4.4.2 Test report.- After the preproduction tests are completed, three complete copies of a test report in accordance with MIL-T-9107 shall be supplied to the procuring activity.

4.4.2.1 Reliability and maintainability information.- The following information shall be submitted as an attachment accompanying the preproduction test report (same quantity and type of copies) or shall

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be included as part of that report:

a. All failures, maintenance, and other events recorded shall be identified by accumulated operating time, miles, cycles, or position in the test procedure as appropriate. Test conditions during the failures or irregular operations identified shall be recorded

b. Description of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified

c. Description of the engineering reasoning verifying corrections made, to be made on production items, or proposed to be made that substantiate the predicted effectiveness of those corrections

d. Test activity or contractor comments on item features or requirements that, if modified, should improve the item

e. Test activity or contractor comments on field conditions or procedures to be avoided or cultivated to increase the reliability and useful life of the item

f. Estimates (±5 percent of actual experienced) of manhours required for each maintenance and servicing action during the tests. A brief description of the qualifications and experience of the personnel involved shall be included and shall be adequate to permit comparison to the personnel anticipated in similar fieldwork.

4.4.3 Preproduction tests.- The preproduction tests shall consist of all the tests described under 4.6.

4.5 Acceptance tests.- The acceptance tests shall consist of the following:

a. Individual test

b. Sampling plan and tests.

4.5.1 Individual test.- Each pumping unit shall be subjected to an examination of product test as described under 4.6.

4.5.2 Sampling plan and tests

4.5.2.1 Lot.- A lot shall consist of pumping units manufactured under essentially the same conditions, of the same material, and submitted for inspection at substantially the same time.

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4.5.2.2 Sampling plan.- Five pumping units shall be selected at random from each lot of 100 or fraction thereof produced and subjected to the following tests as described under 4.6:

a. Performance demonstration

b. Static pressure test.

4.5.2.2.1 Rejection and retest.- When one or more items from a lot fail to meet the specification, acceptance of all items in the lot will be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.2.2 Individual tests may continue.- For production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of the entire lot shall not be made until it is determined that the lot meets all the requirements of the specification.

4.5.3 Defects in items already accepted.- The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product.- The pumping unit shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, and marking. The dimensions, weight, and weight distribution shall be recorded and shall comply with the requirements specified in 3.14 and 3.15.

4.6.2 Performance demonstration.- The performance of the pump shall be demonstrated by pumping CB within the temperature range of -65° to +100° F and by pumping foam liquid within the temperature range of +32° to +125° F until 100 gallons of each agent has been delivered. The pump head shall be maintained at 12 feet of agent. Under these conditions, the pumping unit shall demonstrate ability to deliver agents as specified in 3.6. Equivalent pressures are 10.7 psi and 6.0 psi for 12 feet of CB and foam liquid respectively.

4.6.3 Static pressure.- The pump shall be removed from the assembly. The outlet of the pump shall be plugged and air, at a pressure of 20 psi, shall be applied to the pump inlet. With the pump submerged in water, no appreciable air leakage shall be exhibited over a 15-minute period.

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4.6.4 Initial disassembly examination.- The pumping unit shall be disassembled to the maximum practical extent, and the condition of its parts shall be recorded. Measurements shall be made and recorded with regard to those parts that are subject to wear or distortion during testing.

4.6.5 Dry operation.- The pump shall be dry operated for a period of 1 hour. A dry suction lift of 10 inches of Hg shall be maintained during this period.

4.6.6 Road test.- The pumping unit shall be towed behind a vehicle at speeds up to 15 mph for a total distance of 50 miles. A record shall be kept of any signs of instabilities. Any sign of material failure at the completion of this test shall be cause for rejection.

4.6.7 Environmental tests.- The following environmental tests shall be conducted in accordance with the specified methods of MIL-STD-810. While the specified operating temperature is maintained, the tests specified in 4.6.2 shall be conducted.

a. High temperature - Method 501

b. Low temperature - Method 502

c. Humidity - Method 507

d. Salt fog - Method 509.

4.6.8 Endurance.- The pumping unit shall be operated for a period of 250 hours, 125 hours of which shall be used for pumping CB and the remaining time for pumping foam liquid. The head shall be maintained at 12 feet of the applicable agent. The operation shall be interrupted a minimum of four times with a minimum period of 50 hours between each interruption. The pump shall be drained or pumped dry and allowed to dry out. After pumping foam liquid, the pump may be flushed with water prior to allowing it to dry out. The operation shall begin again and the pump shall demonstrate its self-priming ability.

4.6.8.1 Upon completion of the test specified in 4.6.8, the pump shall successfully complete the test specified in 4.6.3. The pump shall be capable of maintaining a dry suction lift of 10 inches Hg.

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4.6.9 Final disassembly inspection.- Upon completion of all testing, the pumping unit shall be disassembled to the maximum practical extent and examined.

4.6.9.1 Parts subject to wear or distortion shall be measured or otherwise dimensionally inspected and the measurements shall be compared with similar data, compiled prior to the start of the prescribed testing, for determination of wear or distortion. All parts shall be in acceptable condition.

4.6.10 Electromagnetic interference.- The pumping unit shall be tested in accordance with the interference control requirements specified for class IV equipment in MIL-I-26600.

4.6.11 Reliability demonstration and test.- Satisfactory completion of all tests specified herein demonstrates compliance with the quantitative reliability requirements of this specification.

4.6.12 Maintainability verification.- Verification of the maintainability requirements shall be in accordance with MIL-M-26512, appendix B, paragraph entitled Maintenance Task Simulation.

4.7 Inspection of preparation for delivery.- The inspection of the preservation, packaging, packing, and marking shall be in accordance with the instructions of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.- The pumping unit shall be preserved in accordance with MIL-P-116, method I. The electric motor and controls shall be preserved in accordance with MIL-P-116, method II.

5.1.2 Level C.- The pumping unit shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing

5.2.1 Level A.- The pumping unit, preserved and packaged as specified in 5.1.1, shall be packed in exterior-type shipping containers conforming to PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-636, or MIL-B-138. As far as practical, exterior containers shall be of uniform shape and size, of minimum cube and tare consistent with the protection required,

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and contain identical quantities. The gross weight of each pack shall be limited to approximately 200 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or the appendix thereto. Containers shall be provided with a case liner conforming to MIL-L-10547, and shall be sealed in accordance with the appendix thereto. As an alternate method, containers conforming to PPP-B-636 may be sealed at all joints and seams with tape conforming to PPP-T-60.

5.2.2 Level B.- The pumping unit, preserved and packaged as specified in 5.1.1 or 5.1.2, shall be packed in domestic-type exterior containers conforming to PPP-B-585, PPP-B-591, or PPP-B-601. Exterior containers shall be of minimum cube and tare consistent with the protection required. As far as practical, exterior containers shall be of uniform shape and size and shall contain identical quantities. The gross weight of each pack shall be limited to approximately 500 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or the appendix thereto.

5.2.4 Level C.- Packages which require overpacking for acceptance by the carrier shall be packed in exterior-type shipping containers in a manner that will insure safe transportation, at the lowest rate, to the point of delivery. Containers shall meet Consolidated Freight Classification Rules or regulations of other common carriers as applicable to the mode of transportation.

5.3 Physical protection.- Cushioning, blocking, bracing, and bolting as required, shall be in accordance with JAN-P-100, except waterproofing requirements for cushioning materials and containers shall be waived for domestic shipment. The drop tests of JAN-P-100 shall be waived when the item is preserved, packaged, and packed for immediate use or when the drop tests of MIL-P-116 are applicable.

5:4 Marking.- Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

Pump, Rotary, Power Driven, 20 GPM Cap., Two Wheeled Hand Cart, Type MC-1.

6. NOTES

6.1 Intended use.- The Type MC-1 pumping unit is intended for use in the field for transferring CB or foam liquid from drums to crash fire fighting-trucks.

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6.2 Ordering data.- Producrment documents should specify the following:

a. Title, number, and date of this specification

b. Couplings required (see 3.8.2)

c. Number of preproduction samples and the point of inspection

d. Applicable levels of preservation, packaging, and packing.

6.3 Definitions

6.3.1 Since the official nomenclature designates the end items as a pump and one of the components is called a pump, the term "pumping unit" will be used herein where reference is made to the complete end item and the term "pump" will be used only when referring to the component.

6.3.2 Reliability.- Reliability is defined as the probability of performing a special function under given conditions without failure for a specified period of time. Recognizing that, in general, the rate of failure of equipment is fairly constant throughout the life of the equipment, the probability of nonfailure over an operating time interval decreases exponentially as a function of the length of the interval, during which time there is a constant failure rate, and can be expressed as follows:

Confidence level: 90 percent

$$\text{MTBF} = \frac{\text{Total test time}}{2.3}$$

$$\text{Reliability} = e^{-x} \text{ where } x = \frac{\text{Mission time}}{\text{MTBF}}$$

The $2.3 = \frac{4.61}{2}$ is based on the constant for the Poisson/Chi squared distribution, assuming an exponential (2^{e} freedom) distribution even through the failure rate for the test time is zero.

6.3.3 Mean-time-between-failures.- The mean life or mean-time-between-failures is the arithmetical mean (average) of the operating time between failures.

6.3.4 Mission time.- Mission time is the time the pumping unit is operated in transferring CB or foam liquid from drums to crash fire-fighting trucks.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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